2026

Order Delivery & Logistics Performance Analysis using Python, SQL & Power BI

1. Data Dictionary

Field	Description
Order_ID	Unique order identifier
Customer_ID	Unique customer identifier
Order_Date	Date order was placed
Processing_Time_Hours	Order processing time
Shipping_Date	Date order was shipped
Expected_Delivery_Date	Expected delivery date
Actual_Delivery_Date	Actual delivery date
Delivery_Time_Days	Delivery duration
Delivery_Status	On Time / Delayed / Cancelled
Delay_Days	Number of delayed days
Cancellation	Yes / No
Cancellation_Reason	Reason for cancellation
Courier	Delivery company
Warehouse	Fulfilment warehouse
Region	Delivery region
Shipping_Cost	Shipping cost
Order_Value	Order value
Customer_Rating	Customer rating (1–5)
Payment_Method	Payment method
Product_Category	Product category
Priority	Express / Standard

2. README

Project: Order Delivery & Logistics Performance Analysis

Objective: Analyse order processing, delivery delays, cancellations, courier performance, warehouse performance, customer ratings, and order value.

Tools: Python, SQL, Power BI

Dataset: 10,000 orders, 21 columns.

Key results:

- On-Time Rate: **74.72%**
- Delay Rate: **17.16%**
- Cancellation Rate: **8.12%**
- Average Delivery Time: **4.94 days**
- Average Customer Rating: **2.99/5**

SQL & Python File Comments

SQL:

-- Purpose: Validate data and analyse delivery,
-- Cancellation, courier, warehouse and business KPIs.

Python:

Purpose: Clean, validate and analyse the
Order Delivery & Logistics Performance dataset.

Dashboard Documentation

Page 1 – Executive Overview: Overall KPIs and business performance.

Page 2 – Delivery Performance: On-time deliveries, delays and delivery time.

Page 3 – Cancellation Analysis: Cancellation rate and cancellation reasons.

Page 4 – Courier & Warehouse: Compare logistics and warehouse performance.

Page 5 – Customer & Product: Customer ratings, order value and product categories.

Main Measures:

- Total Orders
- Total Order Value
- Total Shipping Cost
- On-Time Rate
- Delay Rate
- Cancellation Rate
- Average Delivery Time
- Average Customer Rating

Portfolio Artefacts

- README
- Data Dictionary
- Cleaned Dataset
- SQL Files
- Python Files
- Power BI Dashboard
- Dashboard Documentation
- Project Summary